

where the asymptotic error analysis (for $h \rightarrow 0$) may be less relevant. Nevertheless there are applications in molecular modelling where higher accuracy is needed in certain parts of a calculation in order to provide optimal sampling. An example will be found in the isokinetic stochastic integrator of Chap. 8.

For more discussion of higher order integrators, refer to [139, 164, 227, 259, 260, 262, 277, 290]. Of particular relevance for molecular dynamics is the excellent numerical study of [153].

3.3.4 The Takahashi-Imada Method

Recall the Takahashi-Imada method introduced in the last chapter,

$$\begin{aligned}\hat{\mathbf{P}} &:= \mathbf{p} - (h/2)\nabla\tilde{U}(\mathbf{q}), \\ \mathbf{Q} &:= \mathbf{q} + h\mathbf{M}^{-1}\hat{\mathbf{P}}, \\ \mathbf{P} &:= \hat{\mathbf{P}} - (h/2)\nabla\tilde{U}(\mathbf{Q}),\end{aligned}$$

with modified potential energy function

$$\tilde{U}(\mathbf{q}) = U(\mathbf{q}) - \frac{h^2}{24}\nabla U(\mathbf{q})^T\mathbf{M}^{-1}\nabla U(\mathbf{q}).$$

One might recognize the modification as being proportional to one part of the commutator expansion in the Verlet method, in fact

$$\nabla U(\mathbf{q})^T\mathbf{M}^{-1}\nabla U(\mathbf{q}) = \{U, \{U, T\}\}.$$

This is certainly not a coincidence. If we replace the potential U by $\tilde{U}$ in the Verlet expansion, we have

$$\begin{aligned}\tilde{H}_h &= T + \tilde{U} + \frac{h^2}{12}\left(\{T, \{T, \tilde{U}\}\} - \frac{1}{2}\{\tilde{U}, \{\tilde{U}, T\}\}\right) + \mathcal{O}(h^4) \\ &= T + U - \frac{h^2}{24}\{U, \{U, T\}\} + \frac{h^2}{12}\left(\{T, \{T, \tilde{U}\}\} - \frac{1}{2}\{\tilde{U}, \{\tilde{U}, T\}\}\right) + \mathcal{O}(h^4) \\ &= H + \frac{h^2}{12}\left(\{T, \{T, \tilde{U}\}\} - \{\tilde{U}, \{\tilde{U}, T\}\}\right) + \mathcal{O}(h^4) \\ &= H + \frac{h^2}{12}\left(\mathbf{p}^T\mathbf{M}^{-1}U''\mathbf{M}^{-1}\mathbf{p} - \nabla U^T\mathbf{M}^{-1}\nabla U\right) + \mathcal{O}(h^4).\end{aligned}$$

Introducing coordinate transformations